Biomedical Signal Processing and Control 24 (2016) 120–127

Contents lists available at ScienceDirect

Biomedical Signal Processing and Control

journal homepage: www.elsevier.com/locate/bspc

Denoising optical coherence tomography using second order total generalized variation decomposition

Jinming Duan[a,*], Wenqi Lu[b], Christopher Tench[c], Irene Gottlob[d], Frank Proudlock[d], Niraj Nilesh Samani[d], Li Bai[a]

[a] School of Computer Science, University of Nottingham, UK
[b] College of Information Engineering, Qingdao University, China
[c] School of Medicine, University of Nottingham, UK
[d] Ophthalmology Department, University of Leicester, UK

ARTICLE INFO

Article history:
Received 11 May 2015
Received in revised form 8 September 2015
Accepted 30 September 2015
Available online 6 November 2015

Keywords:
Optical coherence tomography
Variational methods
Image decomposition
Noise removal
Total generalised variation
Split Bregman algorithm
Fast Fourier transform

ABSTRACT

In this paper, we apply image decomposition for image denoising by considering the speckle noise in the (OCT) image as texture or oscillatory patterns. A novel second order total generalised variation (TGV) decomposition model is proposed to remove noise (texture) from the OCT image. The incorporation of the TGV regularisation in the proposed model can eliminate the staircase side effect in the resulting denoised image (structure). By introducing auxiliary splitting variables and Bregman iterative parameters, a fast Fourier transform based split Bregman algorithm is developed to solve the proposed model explicitly and efficiently. Extensive experiments are conducted on both synthetic and real OCT images to demonstrate that the proposed model outperforms state-of-the-art speckle noise reduction methods.

© 2015 Elsevier Ltd. All rights reserved.

1. Introduction

Optical coherence tomography (OCT) is a non-invasive imaging modality and plays an important role in clinical diagnosis and monitoring of diseases of the retina. However, as with other imaging modalities based on the detection of coherence sources, OCT images are always corrupted by speckle noise, which can mask image features and pose significant challenges for image analysis algorithms [1]. Intensity fluctuation and motion artifact also decrease image quality, and presence of blood vessels in the image makes layer boundaries appear discontinuous. Thus pre-processing is often the first step in OCT image analysis.

There are mainly two classes of methods for speckle noise reduction of OCT images. The first class of methods use physical techniques, such as the frequency compounding and polarization diversity [2–4], to address the noise problem before final image formation. However, these methods require significant modifications to design of the OCT system and can be too complicated and expensive to apply in practise. The second class of methods rely on post-processing of images and representatives of this class include filter based and diffusion based approaches.

Adaptive adaptive filters calculate the value of each pixel using the information in the local neighbourhood of the pixel. These include hybrid median filter [5], homomorphic Wiener filter [6], enhanced Lee filter [7], symmetric nearest neighbour filter [8], Kuwahara filter [9], etc. Wavelet-based filters have the advantage of denoising on multiscale resolutions and are more desirable for dealing with correlated noise. Ozcan et al. [10] compared standard digital denoising methods and concluded that wavelet thresholding with the shift invariant wavelets yielded the best results. A spatially adaptive wavelet filter was described in [11], where a set of statistical wavelet coefficients were applied to estimate the thresholds for denoising OCT images. Mayer et al. [12] proposed a wavelet denoising algorithm for OCT images. The wavelet coefficients are first calculated based on local noise and structure estimation, and then weighted, averaged and used to reconstruct the denoised image. Puvanathasan et al. [13] proposed a type-II fuzzy thresholding algorithm for choosing wavelet filtering thresholds. Modern wavelets such as the dual tree complex wavelet [14] and curvelet transform [15] have also been employed to remove speckle noise from the

* Corresponding author. Tel.: +44 7591848541.
E-mail address: psxjd3@nottingham.ac.uk (J. Duan).

http://dx.doi.org/10.1016/j.bspc.2015.09.012
1746-8094/© 2015 Elsevier Ltd. All rights reserved.

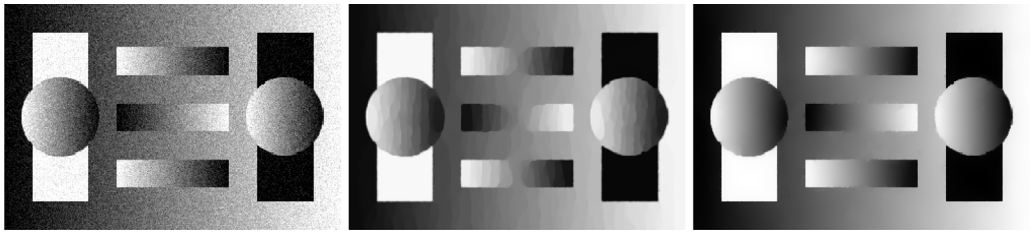


Fig. 1. Denoising a synthetic image with TV and TGV penalty. The noisy test image f (left), its TV regularisation (middle) and its TGV (right), which does not have the staircase effect.

Table 1
Methods and their abbreviations for comparison.

No.	Full name	Abbreviations
1	Hybrid median filter	HMF
2	Haar wavelet (hard thresholding)	HWT
3	Dual tree complex wavelet (soft thresholding)	DTCWT
4	Nonlinear complex diffusion filter	NCDF
5	Anisotropic coherent enhancing diffusion	ACED
6	Vese–Osher decomposition model	VO

OCT images which have produced better results than conventional wavelets.

Diffusion based approach can smooth images while preserving edges of objects and it often leads to partial differential equations (PDEs). Salinas et al. [16] compared the performance of Perona–Malik isotropic diffusion with a nonlinear complex diffusion filter for OCT denoising [17]. The nonlinear complex diffusion was combined with the anisotropic coherence enhancing diffusion [18] for noise reduction, segmentation and structure analysis in retinal OCT images. Bernardes et al. [19] proposed a new formulation of the nonlinear complex diffusion filter. One particular advantage of their method over the original nonlinear complex diffusion is that the conductance parameter k in their formulation can be adaptive to the OCT data automatically. Bernardes' method was then extended to 3D OCT image denoising with the GPU programming by Rodrigues et al. [20]. In addition, Puvanathasan et al. [21] presented the application of the interval type-II fuzzy anisotropic diffusion algorithm, where they considered the uncertainty in the calculated diffusion coefficient and appropriate adjustments to the coefficient were made. They also included edge information and a noise estimate in their formulation of the nonlinear isotropic diffusion for OCT speckle reduction.

Variational models offer powerful processing capabilities for imaging [22–27]. They have been widely used in the last two decades but have only been used recently for OCT image processing [23,28]. In this paper, based on the classical Vese–Osher (VO)

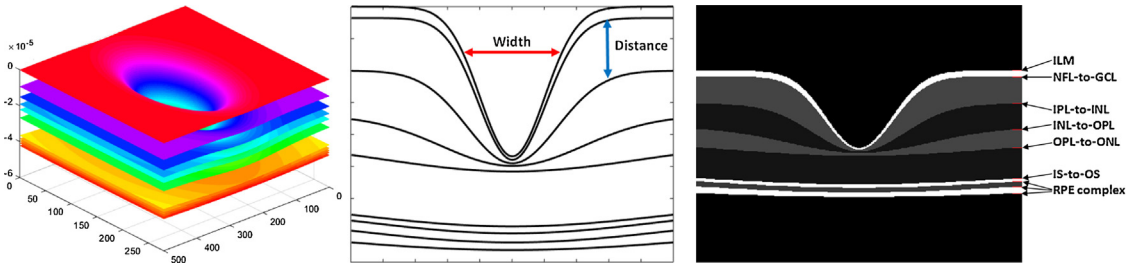


Fig. 2. Procedure of simulation for the synthetic OCT image. Left image: 3D sharp generated by Eq. (4.1); Middle image: The middle cross of the 3D sharp; Right image: Synthetic OCT image (B-scan) with 9 retinal layers. They are respectively ILM (internal limiting membrane), NFL (nerve fiber layer), GCL (ganglion cell layer), IPL (inner plexiform layer), INL (inner nuclear layer), OPL (outer plexiform layer), ONL (outer nuclear layer), IS (inner segment), OS (outer segment), and RPE (retinal pigment epithelium).

Table 2
Comparison of the PSNR, SNR, RMSE and SSIM using the different methods on the synthetic OCT image (i.e. Fig. 2 right) with 5 different noise variances.

	PSNR test					SNR test				
Noise variance	0.02	0.04	0.06	0.08	0.1	0.02	0.04	0.06	0.08	0.1
Noisy image	17.5325	14.9554	13.5430	12.5612	11.8256	8.4672	5.8901	4.7777	3.4959	2.7603
HMF	21.0741	20.2947	19.9165	19.4774	19.0066	12.0088	11.2294	10.8512	10.4121	9.9413
HWT	24.3981	22.1582	20.8430	19.3209	18.0995	15.3328	13.0929	11.7777	10.2556	9.0342
DTCWT	27.8702	24.7867	23.1115	21.9070	20.4468	18.8049	15.7214	14.0462	12.8417	11.3815
NCDF	20.9252	19.0848	18.4835	17.3998	16.8213	11.8599	10.0195	9.4182	8.3345	7.7560
ACED	28.1206	26.2074	24.6428	23.0592	21.4316	19.0553	17.1421	15.5775	13.9939	12.3664
VO	28.2919	26.4631	24.9032	23.5448	22.3510	19.2266	17.3978	15.8379	14.4795	13.2857
Ours	**32.0284**	**29.5787**	**27.4901**	**25.9198**	**24.4470**	**22.9631**	**20.5134**	**18.4248**	**16.8545**	**15.3817**
	RMSE test					SSIM test				
Noise variance	0.02	0.04	0.06	0.08	0.1	0.02	0.04	0.06	0.08	0.1
Noisy image	9.9006	10.3206	10.5214	10.6182	10.6609	0.1327	0.0978	0.0826	0.0730	0.0655
HMF	6.0263	6.5034	6.8127	7.0316	7.0406	0.6844	0.5745	0.4979	0.4496	0.4133
HWT	4.8105	4.9782	5.0147	5.0320	5.0487	0.9249	0.8923	0.8709	0.8369	0.8273
DTCWT	4.1782	4.4722	4.5374	4.5669	4.5834	0.9317	0.9094	0.8934	0.8730	0.8513
NCDF	4.7767	4.8449	4.8746	4.9408	5.0166	0.7998	0.7808	0.7781	0.7768	0.7722
ACED	5.0821	5.4730	5.4769	5.5201	5.5256	0.7679	0.6761	0.6016	0.5141	0.4654
VO	4.0315	4.0812	4.0943	4.0973	4.1099	0.9728	0.9624	0.9517	0.9394	0.9267
Ours	**3.7756**	**3.9375**	**3.9437**	**3.9456**	**3.9759**	**0.9871**	**0.9781**	**0.9684**	**0.9590**	**0.9481**

decomposition model, we propose a novel second order total generalised variation (TGV) decomposition model decomposing an OCT image into a structure image and a texture image, where the former is the denoised image whilst the latter contains the speckle noise in the original image. Unlike original VO model, the proposed method would not lead to the staircase effect (i.e. an undesirable jagged appearance) in the denoised image due to the TGV regularisation. In order to avoid directly calculating the high order nonlinear partial differential equations of the proposed model, the standard split Bregman algorithm [29] is modified by introducing three auxiliary splitting variables and three Bregman iterative parameters. Then, the original problem can be transformed into a multivariate optimisation problem. The analytical fast Fourier transform (FFT) [26,27,30] and generalised soft thresholding equations [31] can be elegantly applied to solve each variable efficiently without any iteration.

The contributions of the paper are twofold. First, we propose an second order total generalised variation decomposition model that can remove the noise from the OCT image. The resulting denoised part does not have the undesirable staircase effect associated with the original VO model. Second, we develop the FFT-based split Bregman algorithm to solve the proposed model explicitly and efficiently.

The paper is organised as follows: Section 2 introduces the proposed second order TGV decomposition model for OCT image denoising; Section 3 describes the split Bregman algorithm for solving the proposed variational model efficiently; Section 4 documents the experiments to validate the proposed method on both synthetic and real OCT images. A novel method to synthesise OCT images is introduced. Section 5 concludes the paper.

2. Methodology

2.1. The total generalised variation model

In this section, the concept of total generalised variation (TGV) [25] is introduced. TGV is based on the total variation (TV) model. For a given image u, the TV model is denoted as

$$\mathrm{TV}(u) = \int_\Omega |\nabla u| dx \tag{2.1}$$

The second order TGV can be represented as

$$\mathrm{TGV}^2_\alpha(u) = \min_p \left\{ \alpha_1 \int_\Omega |\nabla u - p| + \alpha_0 \int_\Omega |\varepsilon(p)| \right\} \tag{2.2}$$

where $\varepsilon(p) = (\nabla p + \nabla p^{\mathrm{T}})/2$ is the symmetrised derivative and the minimum of (2.2) is taken over all the symmetrised gradient of the deformation field $p = (p_1, p_2)$ on the open subset Ω. The positive penalty parameters α_0 and α_1 balance the first and second derivatives respectively .

The main property of TGV regularisation is its capability of representing image characteristics up to an arbitrary order of differentiation (piecewise constant, piecewise affine, piecewise quadratic, etc.), resulting in good image reconstruction results. The TGV functional is convex, which enables a global optimal solution to be found. In contrast to TV, TGV takes both first and second derivatives into accounts, thus highlighting objects edges while smoothing regions in the image. Therefore, using TGV as a regulariser can eliminate the staircase effect which is usually associated with TV regulariser. Finally, TGV is translation invariant and rotationally invariant. Example results by TV and TGV are demonstrated in Fig. 1.

2.2. The proposed TGV decomposition model for OCT denoising

Speckle noise in OCT images can be approximated as a multiplicative noise:

$$f = u\eta + n \tag{2.3}$$

where u represents the noise free OCT image; f is the noise observation of u; η and n are the multiplicative speckle and additive noise, respectively. The additive noise component can be ignored because it is significantly small compared with the multiplicative speckle noise [32]. Thus, speckle noise in OCT images can be modelled as

$$f = u\eta \tag{2.4}$$

According to statistical optics, speckle noise η in OCT images is multiplicative noise and can be modelled by Rayleigh distribution [2,3]. However, due to the limited dynamic range of display monitors, the OCT signal is normally log-compressed to fit the display range. The compression changes the characteristics of the Rayleigh probability density function [2,33]. Thus, prior to the logarithm compression, speckle noise can be approximated as multiplicative noise and afterwards additive noise. The OCT denoising problem thus becomes removing additive speckle noise, which can be treated as oscillating texture and removed using the following VO model for image decomposition:

$$E(u, g) = \frac{1}{2} \int_\Omega (f - u - div(g))^2 + \gamma \int_\Omega |g| + \alpha \int_\Omega |\nabla u| \tag{2.5}$$

Numerical experiments [24] showed that the VO model outperforms the classical ROF model [34] in decomposing oscillating texture from images. The drawbacks of the VO model is that the resulting structure image u can produce staircase effect due to the use of TV regularisation term. In order to remedy the side effect, we apply the TGV regularisation in (2.2) and propose the following TGV decomposition model

$$\min_{g,u} \left\{ \frac{1}{2} \int_\Omega (f - u - div(g))^2 + \gamma \int_\Omega |g| + \mathrm{TGV}^2_\alpha(u) \right\} \tag{2.6}$$

In the first energy term of (2.6), f is the original noisy OCT image after logarithmic transformation; u is a piecewise smooth function representing the denoised image; $div(g)$ is the divergence function, with the vector field $g = (g_1, g_2)$, representing speckle noise when (2.6) converges, that is, $f - u \approx div(g)$. In the second energy term of (2.6), the L^1 norm of vector g allows to capture the oscillations of function $div(g)$ in energy minimisation – a highly oscillating function often has a small L^1 norm. γ is a positive penalty parameter, controlling how many oscillations will appear in the resulting noise image $div(g)$. The last TGV regularisation term on u is able to preserve edges of objects while smoothing the OCT image without yielding the staircase effect.

2.3. Numerical implementation

In order to implement numerical algorithm on a digital image, the discrete setting of the proposed method [25] should be considered. For discretisation, we apply a two dimensional regular Cartesian grid of size $M \times N$ and spatial mesh size $h > 0$. We will use the superscript h to indicate the discrete setting. The differential operators ε^h, ∇^h, div^h are approximated using the first order finite differences. The detailed implementation of these operators are given below.

$$\varepsilon^h(p) = \begin{pmatrix} \partial^h_x p_1, \left(\partial^h_y p_1 + \partial^h_x p_2\right)/2 \\ \left(\partial^h_y p_1 + \partial^h_x p_2\right)/2, \partial^h_y p_2 \end{pmatrix} \tag{2.7}$$

$$\nabla^h u = \left(\partial_x^h u, \partial_y^h u\right) \tag{2.8}$$

$$div^h(g) = \partial_x^h g_1 + \partial_y^h g_2 \tag{2.9}$$

where ∂_x^h and ∂_y^h are two discrete first order derivatives. The coordinates x and y are oriented along columns and rows of the image, respectively. With $U^h = C^{MN}$ and $G^h = \left(C^{MN}\right)^2$, the discrete version of the decomposition problem (2.6) can be written as

$$\min_{u\in U^h, g^h\in G^h}\left\{\frac{1}{2}\left\|f^h - u^h - div^h\left(g^h\right)\right\|_2^2 + \gamma\left\|g^h\right\|_1 + \mathrm{TGV}_\alpha^2\left(u^h\right)\right\} \tag{2.10}$$

where $f^h \in U^h$ is the log-compressed noisy input data. $\left\|g^h\right\|_1$ is the discrete L^1 norm of $|g|$ and $\mathrm{TGV}_\alpha^2\left(u^h\right)$ is the discrete total generalised variation. According to (2.2) and using the corresponding discrete operators, the discretised problem (2.10) then becomes

$$\min_{u\in U^h, g^h\in G^h, p^h\in P^h}\left\{\frac{1}{2}\left\|f^h - u^h - div^h\left(g^h\right)\right\|_2^2 + \gamma\left\|g^h\right\|_1 + \alpha_1\left\|\nabla^h u^h - p^h\right\|_1 + \alpha_0\left\|\varepsilon^h\left(p^h\right)\right\|_1\right\} \tag{2.11}$$

where $P^h = \left(C^{MN}\right)^2$. Note that only first order differential operators are used in this formulation. There are totally three L^1 norms in the proposed method. The recent split Bregman method [29] is one of the most efficient numerical schemes for solving the L^1-based variational models [35,36]. It is easy to implement, has a low memory requirement, and is suitable for parallel implementation. The main idea of the algorithm is to introduce several auxiliary variables and Bregman iterative parameters, with the aim to transform the original energy functional into several simpler linear subproblems. Then the fast Fourier transform (FFT) and analytical generalised soft thresholding equation can be efficiently applied to the subproblems. It is worth mentioning here the augment Lagrangian method is equivalent to the split Bregman method. However, the split Bregman contains fewer energy terms, leading to a simpler implementation. Therefore, it is sensible to use the split Bregman method to optimise the proposed model (2.11). Details of this algorithm are introduced in the next section.

3. The split Bregman method for the proposed TGV model

First, we introduce the auxiliary variables $\left(m^h, w^h, v^h, \tau^h, \xi^h, \kappa^h\right)$ and positive penalty parameters $\left(\theta_1, \theta_2, \theta_3\right)$, transforming the functional in the minimisation problem (2.11) into the following form

$$\begin{aligned} E\left(u^h, g^h, p^h, v^h, m^h, w^h; \tau^h, \xi^h, \kappa^h\right) &= \frac{1}{2}\left\|f^h - u^h - div^h\left(g^h\right)\right\|_2^2 \\ &+ \gamma\left\|m^h\right\|_1 + \frac{\theta_1}{2}\left\|m^h - g^h - \tau^h\right\|_2^2 + \alpha_1\left\|w^h\right\|_1 + \frac{\theta_2}{2}\left\|w^h\right. \\ &\left. - \nabla^h u^h + p^h - \xi^h\right\|_2^2 + \alpha_0\left\|v^h\right\|_1 + \frac{\theta_3}{2}\left\|v^h - \varepsilon^h\left(p^h\right) - \kappa^h\right\|_2^2 \end{aligned} \tag{3.1}$$

The primal and dual variables in (3.1) are:

- $u^h \in C^{MN}$ represents the noise free image we aim to find;
- $g^h = \left(g_1^h, g_2^h\right) \in \left(C^{MN}\right)^2$ is a 2D vector valued function whose divergence represents speckle noise;
- $p^h = \left(p_1^h, p_2^h\right) \in \left(C^{MN}\right)^2$ is the symmetrised gradient of the deformation space;
- $m^h = \left(m_1^h, m_2^h\right) \in \left(C^{MN}\right)^2$ is a 2D vector valued function related to the vector field g^h;
- $w^h = \left(w_1^h, w_2^h\right) \in \left(C^{MN}\right)^2$ is a 2D vector valued function related to the gradient function $\nabla^h u^h - p^h$;
- $v^h = \begin{pmatrix} v_{11}^h, v_3^h \\ v_3^h, v_{22}^h \end{pmatrix} \in \left(C^{MN}\right)^4$ is a 4D vector valued function related to the symmetrised derivative $\varepsilon^h\left(p^h\right)$;
- $\tau^h = \left(\tau_1^h, \tau_2^h\right) \in \left(C^{MN}\right)^2$ is the Bregman iterative parameter for enforcing $m^h = g^h$;
- $\xi^h = \left(\xi_1^h, \xi_2^h\right) \in \left(C^{MN}\right)^2$ is the Bregman iterative parameter for enforcing $w^h = \nabla^h u^h - p^h$;
- $\kappa^h = \begin{pmatrix} \kappa_{11}^h, \kappa_3^h \\ \kappa_3^h, \kappa_{22}^h \end{pmatrix} \in \left(C^{MN}\right)^4$ is the Bregman iterative parameter for enforcing $v^h = \varepsilon^h\left(p^h\right)$.

In order to find all the minimisers for $u^h \in U^h, g^h \in G^h$ and $p^h \in P^h$ in the minimization problem (2.11), an alternating optimization method is used to search all the saddle points of functional (3.1). Specifically, the variables $\left(g^h, p^h, v^h, m^h, w^h; \tau^h, \xi^h, \kappa^h\right)$ are fixed first to optimise u^h by employing the following FFT:

$$u^h = \Re\left(\mathcal{F}^{-1}\left(\frac{\mathcal{F}\left(f^h - div^h\left(g^h\right) - \theta_2 div^h\left(w^h + p^h - \xi^h\right)\right)}{1 - \theta_2\mathcal{F}\left(\Delta^h\right)}\right)\right) \tag{3.2}$$

where F^{-1} denotes the discrete inverse Fourier transform; $\Re$ is the real part of a complex number; $\Delta^h = \partial_x^h\partial_x^h + \partial_y^h\partial_y^h$ is the discrete second order Laplace operator; and "–" stands for point-wise division of matrices.

Next, the variables $\left(u^h, p^h, v^h, m^h, w^h; \tau^h, \xi^h, \kappa^h\right)$ are fixed. Using the FFT, the analytical forms with respect to the vector $g^h = \left(g_1^h, g_2^h\right)$ can be derived as follows:

$$g_1^h = \Re\left(\mathcal{F}^{-1}\left(\frac{a_{22}\mathcal{F}(h_1) - a_{12}\mathcal{F}(h_2)}{D}\right)\right) \tag{3.3}$$

$$g_2^h = \Re\left(\mathcal{F}^{-1}\left(\frac{-a_{21}\mathcal{F}(h_1) + a_{11}\mathcal{F}(h_2)}{D}\right)\right) \tag{3.4}$$

where $h_1 = \partial_x^+\left(u^h - f^h\right) - \theta_1\left(\tau_1^h - m_1^h\right)$ $h_2 = \partial_y^h\left(u^h - f^h\right) - \theta_1\left(\tau_2^h - m_2^h\right)$ and D is the determinant of the following coefficient matrix

$$\begin{pmatrix} a_{11}, a_{12} \\ a_{21}, a_{22} \end{pmatrix} = \begin{pmatrix} \mathcal{F}\left(\theta_1 - \partial_x^h\partial_x^h\right), \mathcal{F}\left(-\partial_x^h\partial_y^h\right) \\ \mathcal{F}\left(-\partial_y^h\partial_x^h\right), \mathcal{F}\left(\theta_1 - \partial_y^h\partial_y^h\right) \end{pmatrix}$$

Then, $\left(u^h, g^h, v^h, m^h, w^h; \tau^h, \xi^h, \kappa^h\right)$ are fixed, and the minimisation problem with respect to the deformation field $p^h = \left(p_1^h, p_2^h\right)$ can be analytically solved through the FFT under the assumption of the circulant boundary condition:

$$p_1^h = \Re\left(\mathcal{F}^{-1}\left(\frac{b_{22}\mathcal{F}(d_1) - b_{12}\mathcal{F}(d_2)}{B}\right)\right) \tag{3.5}$$

$$p_2^h = \Re\left(F^{-1}\left(\frac{-b_{21}\mathcal{F}(d_1) + b_{11}\mathcal{F}(d_2)}{B}\right)\right) \tag{3.6}$$

where

$$d_1 = \theta_2\left(\partial_x^h u^h + \xi_1^h - w_1^h\right) - \theta_3\partial_x^h\left(v_{11}^h - \kappa_{11}^h\right) - \theta_3\partial_y^h\left(v_3^h - \kappa_3^h\right)$$

$$d_2 = \theta_2\left(\partial_y^h u^h + \xi_2^h - w_2^h\right) - \theta_3\partial_y^h\left(v_{22}^h - \kappa_{22}^h\right) - \theta_3\partial_x^h\left(v_3^h - \kappa_3^h\right)$$

B is the determinant of the following coefficient matrix

$$\begin{pmatrix} b_{11}, b_{12} \\ b_{21}, b_{22} \end{pmatrix} = \begin{pmatrix} \mathcal{F}\left(\theta_2 - \theta_3 \partial_x^h \partial_x^h - \frac{\theta_3}{2} \partial_y^h \partial_y^h\right), \mathcal{F}\left(-\frac{\theta_3}{2} \partial_y^h \partial_x^h\right) \\ \mathcal{F}\left(-\frac{\theta_3}{2} \partial_x^h \partial_y^h\right), \mathcal{F}\left(\theta_2 - \frac{\theta_3}{2} \partial_x^h \partial_x^h - \theta_3 \partial_y^h \partial_y^h\right) \end{pmatrix}$$

The optimisation problem with respect to v^h can be solved component-wisely through the following 4-dimensional analytical soft generalized with the convention that $0 \cdot (0/0) = 0$

$$v^h = \max\left(\left\|\varepsilon^h\left(p^h\right) + \kappa^h\right\|_2 - \frac{\alpha_0}{\theta_3}, 0\right) \frac{\varepsilon^h\left(p^h\right) + \kappa^h}{\left\|\varepsilon^h\left(p^h\right) + \kappa^h\right\|_2} \tag{3.7}$$

The m^h and w^h problems are given by following two 2-dimensional analytical soft generalized thresholding equations, respectively

$$m^h = \max\left(\left\|g^h + \tau^h\right\|_2 - \frac{\gamma}{\theta_1}, 0\right) \frac{g^h + \tau^h}{\left\|g^h + \tau^h\right\|_2} \tag{3.8}$$

$$w^h = \max\left(\left\|\nabla^h u^h - p^h + \xi^h\right\|_2 - \frac{\alpha_1}{\theta_2}, 0\right) \frac{\nabla^h u^h - p^h + \xi^h}{\left\|\nabla^h u^h - p^h + \xi^h\right\|_2} \tag{3.9}$$

The Bregman iterative parameters $\left(\tau^h, \xi^h, \kappa^h\right)$ are updated as shown from Step 10 to 12 in Algorithm 1. From Step 10 to 12, the Bregman iterative parameter on the right-hand side is the one in the previous iteration. The formulations (3.2)–(3.9) are all analytical without any iteration. Thus, the algorithm is simple and efficient. In addition, the algorithm can be implemented on parallel hardware such as the graphics processing units to further speed up the process. The stopping criterion is chosen as the relative errors of u^h is smaller than 0.001 and in particular the spatial mesh size $h = 1$.

Algorithm 1. Split Bregman algorithm for TGV decomposition model (2.11).

01:	***function*** $\mathrm{TGV}_{Decomposition}(f)$
02:	Initialization: Set $\left(g^h, p^h, v^h, m^h, w^h; \tau^h, \xi^h, \kappa^h\right) = 0$, $\left(\gamma, \alpha_0, \alpha_1, \theta_1, \theta_2, , \theta_3\right) > 0$
03:	***repeat***
04:	Compute u^h according to (3.2)
05:	Compute $g^h = \left(g_1^h, g_2^h\right)$ according to (3.3) and (3.4)
06:	Compute $p^h = \left(p_1^h, p_2^h\right)$ according to (3.5) and (3.6)
07:	Compute v^h according to (3.7)
08:	Compute m^h according to (3.8)
09:	Compute w^h according to (3.9)
10:	Update Bregman iterative parameter $\tau^h \leftarrow \tau^h + g^h - m^h$
11:	Update Bregman iterative parameter $\xi^h \leftarrow \xi^h + \nabla^h u^h - p^h - w^h$
12:	Update Bregman iterative parameter $\kappa^h \leftarrow \kappa^h + \varepsilon\left(p^h\right) - v^h$
13:	***until*** convergence of u
14:	***return*** u
15:	***end function***

4. Numerical experiments

The proposed algorithm is compared with six state-of-the-art speckle noise reduction algorithms on both synthetic and real data. These algorithms are the hybrid median filter (HMF) [5], Haar wavelet (HWT) using hard thresholding, dual tree complex wavelet (DTCWT) using soft thresholding [14], nonlinear complex diffusion filter (NCF) [16], anisotropic coherent enhancing diffusion (ACED) [18], and Vese–Osher decomposition model [24]. The methods and their abbreviations are listed in Table 1. For the synthetic OCT data, the metrics for quantitative evaluation are the peak signal-to-noise ratio (PSNR), signal-to-noise ratio (SNR), root mean square error (RMSE), and structure similarity index map (SSIM). For the real human eye data, visual inspection is used for qualitative evaluation. All the experiments are performed using Matlab 2014b on a Windows 7 platform with an Intel Xeon CPU E5-1620 at 3.70 GHz and 32 GB memory.

4.1. Synthetic OCT images

To synthesise OCT images for quantitative evaluation of the proposed methods, a Gaussian function is used to generate noise free OCT images with arbitrary number of retinal layers:

$$f(x_i, y_i) = -\frac{a_i}{\sqrt{2\pi\sigma_i^2}} \exp\left(-\frac{x_i^2 + y_i^2}{2\sigma_i^2}\right) - b_i \tag{4.1}$$

where the subscript i determines the number of retinal layers; σ_i is standard deviation, together with coefficient a_i, controlling the width of the ith layer (indicated by a red arrow in the middle image of Fig. 2); b_i sets the width (indicated by a blue arrow in the middle image of Fig. 2) of the ith layer. By choosing appropriate parameters, the 3D shape of function (4.1) is obtained as shown in Fig. 2 left, where $i = 9$ representing a total of 9 retinal layers. The middle cross-section of the 3D shape is shown in Fig. 2 middle.

Based on the middle image in Fig. 2 and the knowledge on the reflectance of different retinal layers (e.g. the ILM layer should be one of the most hyper-reflective layers in a retinal OCT image), the synthetic noise free OCT image can be generated as shown in the last image of Fig. 2. As mentioned in Section 2, OCT image denoising boils down to removing additive speckle noise in the image. In order to quantitatively compare the different algorithms on the noisy OCT images with varying degrees of additive speckle noise, the clean synthetic image is corrupted by additive Gaussian noise with zero-mean and different variances. Five variance values from 0.02 to 0.1 at 0.02 intervals are used.

Table 2 contains the four evaluation metrics for quantitative comparison of different methods on the synthetic OCT image corrupted with 5 different noise variances. In order to make the data in Table 2 clearer, we present Fig. 3 for better analysis. For each of the 5 experiments, Fig. 3 plots of the PSNR, SNR, RMSE and SSIM against the increasing levels of noise. As can be seen, NCDF performs the worst in Fig. 3(a) and (b), whilst HMF performs the worst in Fig. 3(c) and (d). For wavelet denoising, modern DTCWT performs better than the conventional HWT. The variational VO model is the second best among all the methods, while the proposed method performs the best in all cases.

In Fig. 4, one denoising case (i.e. noise variance 0.02) in Table 2 is given for visual comparison. As can be seen, the traditional adaptive HMF seems to perform the worst. The resolution of image by HWT using hard thresholding tends to be coarser, while the retinal layers by DTCWT and NCDF methods become blurred. Though ACED can enhance the directional structures, it cannot remove noise entirely, observed in Fig. 4(f). From Fig. 4(g) and (h), one can see both VO model and our proposed model can remove speckle noise while well preserve the edges of the retinal layers. However, VO model uses the first order total variation regularisation and slightly leads to the staircase effect.

4.2. Real data

Three more experiments on the real OCT images are used to further evaluate the different methods against the existing denoising methods, which are presented from Figs. 5–7. The real OCT data was imaged from the left and right eye of 15 healthy adults (mean age = 39.8 years, SD = 8.6 years; 7 male, 8 female) using a spectral domain OCT device (ENVISU C class 2300, Bioptigen, axial resolution = 3.3 μm, scan depth = 3.4 mm, 32,000 A-scans per second)

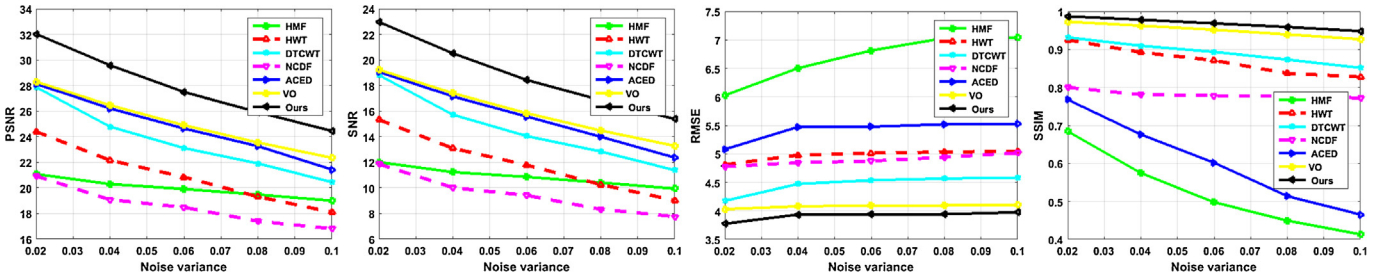


Fig. 3. Performance plots of the different methods applied to the synthetic data with different noise variances. From left to right is respectively the PSNR index, SNR index, RMSE index and SSIM index.

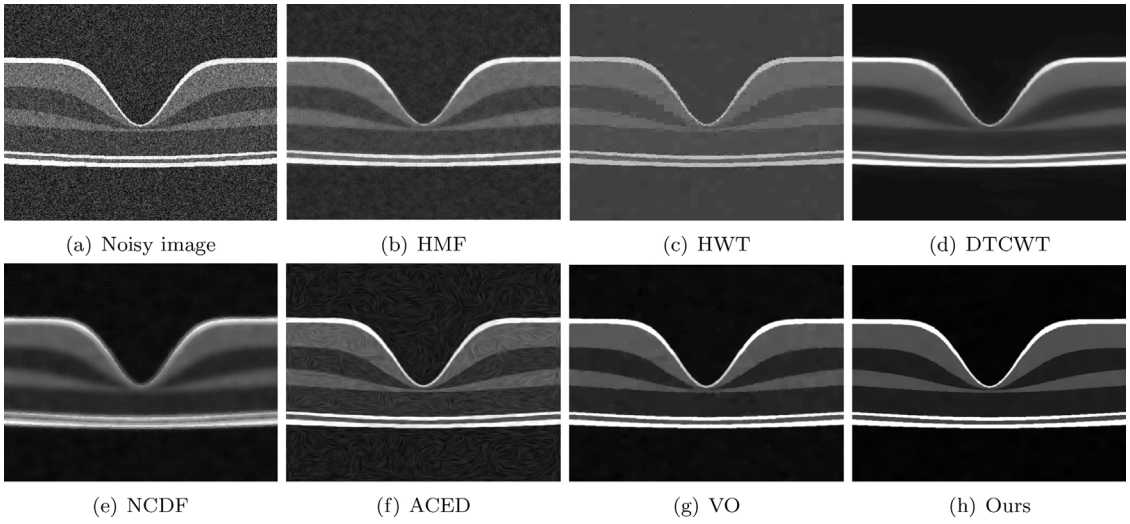


Fig. 4. Performance comparison of the different methods on the synthetic noisy OCT image (a).

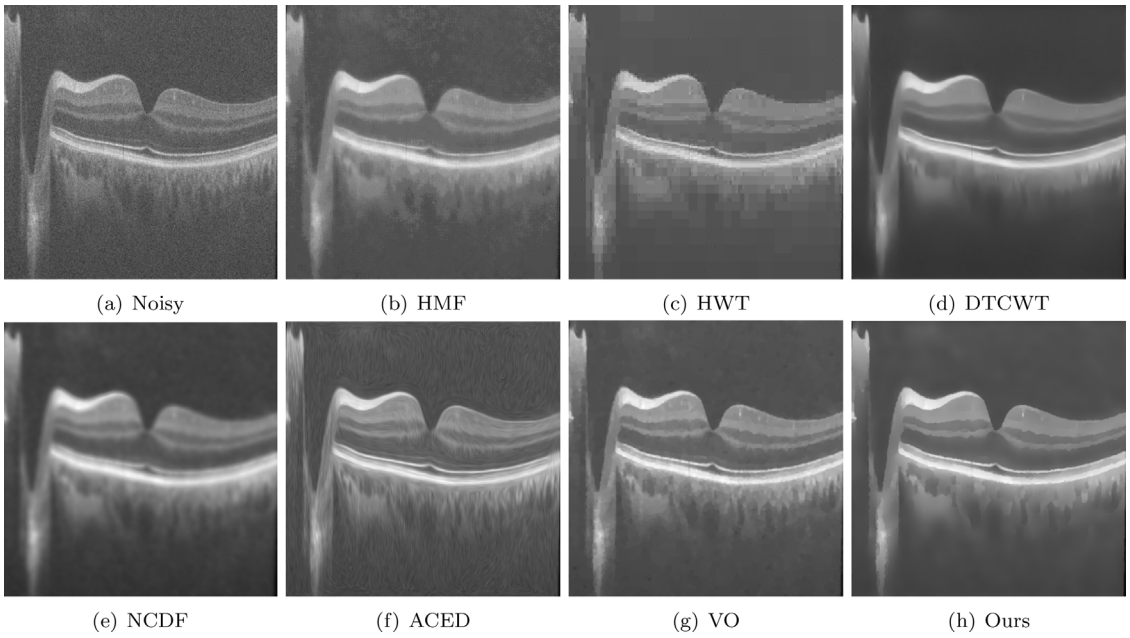


Fig. 5. Performance comparison of the different methods on the real noisy OCT image Fig. 5(a).

with a chin rest to stabilise the head. A 5 mm × 10 mm volumetric scan sequence was used consisting of 500 A-scans × 100 horizontal B-scans. The B-scan located at the foveal centre was identified from the lowest point in the foveal pit where the cone outer segments where elongated (indicating cone specialisation). A good method should significantly suppress the noise within the OCT images while keep the retinal layers continuous and complete.

From these three experiments, it is obvious to see that HMF method produces a large amount of artefacts, HWT leads to the mosaic appearance, DTCWT oversmooths the image, and NCDF

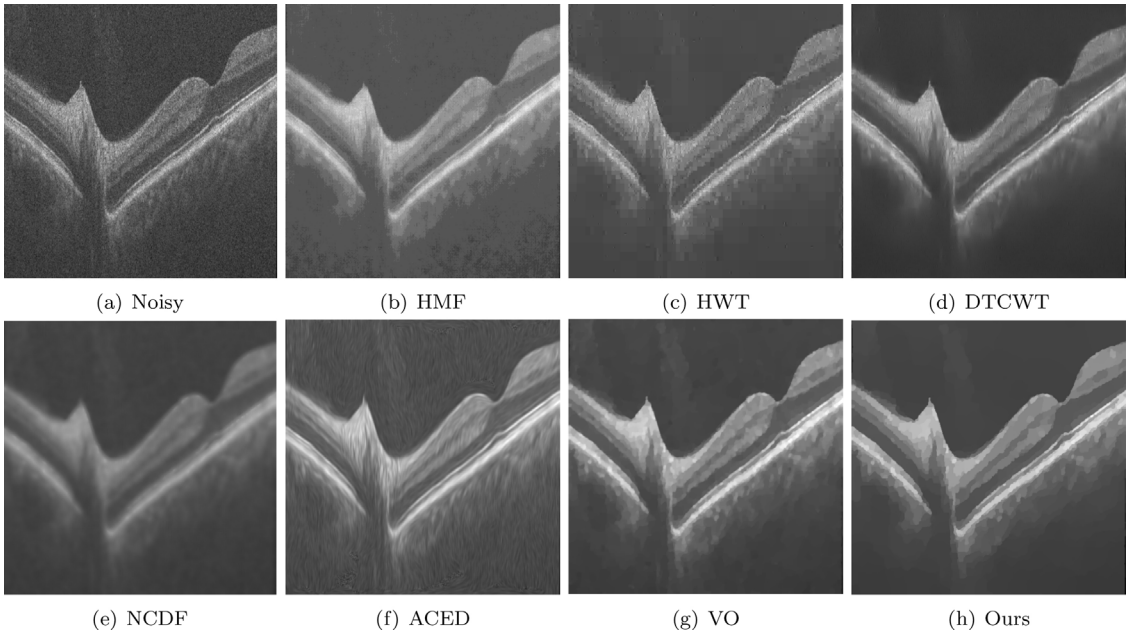


Fig. 6. Performance comparison of the different methods on the noisy OCT image Fig. 6(a).

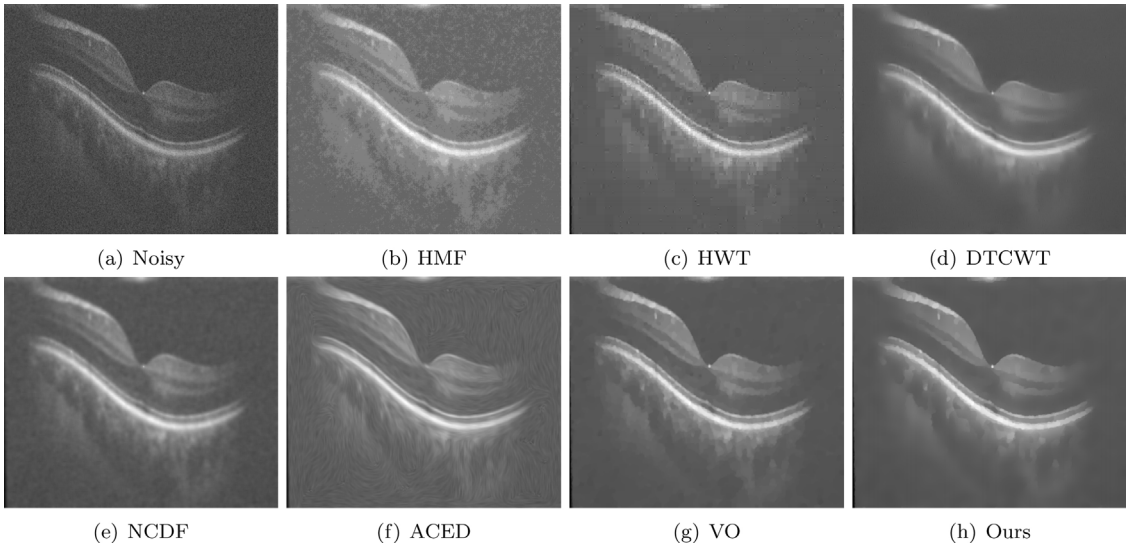


Fig. 7. Performance comparison of the different methods on the noisy OCT image Fig. 7(a).

blurs the edges of the retinal layers. The result by ACED displays the strong directional characteristics but still looks quite noisy. VO model yields more pleasant denoised results but the undesirable staircase effect appears. As can be seen from Figs. 6(g) and 7(g), when inhomogeneous intensities are increasing, the staircase becomes more obvious. The proposed method significantly reduces noise, well preserves the details of the retinal structure and totally eliminates the staircase effect.

5. Conclusion

In this paper, a new second order total generalised variation model is introduced for image decomposition to remove speckle noise from OCT images. The advantages of the proposed model also include computational efficiency through the use of the explicit FFT-based split Bregman algorithm.

A quantitative evaluation on the synthetic OCT data shows that the proposed method is capable of suppressing noise better than the existing state-of-the-art speckle reduction methods. A visual inspection of the denoised images from in-vivo human retina scans shows that the method reduces noise effectively without degrading boundaries of retinal layers or generating staircase effects.

Future application will involve the segmentation of retinal layers in the OCT images preprocessed by the proposed method for analysis tools for ophthalmologists, as it can effectively suppress the noise in the images without losing much retinal structural information.

References

[1] W. Drexler, J.G. Fujimoto, Optical Coherence Tomography: Technology and Applications, Springer Science & Business Media, 2008.
[2] J.M. Schmitt, S.H. Xiang, K.M. Yung, Speckle in optical coherence tomography, J. Biomed. Opt. 4 (1) (1999) 95–105.

[3] M. Pircher, E. Go, R. Leitgeb, A.F. Fercher, Christoph K. Hitzenberger, et al., Speckle reduction in optical coherence tomography by frequency compounding, Journal of Biomedical Optics 8 (3) (2003) 565–569.
[4] A.E. Desjardins, B.J. Vakoc, W.Y. Oh, S.M. Motaghiannezam, G.J. Tearney, B.E. Bouma, Angle-resolved optical coherence tomography with sequential angular selectivity for speckle reduction, Opt. Express 15 (10) (2007) 6200–6209.
[5] A. Nieminen, P. Heinonen, Y. Neuvo, A new class of detail-preserving filters for image processing, IEEE Trans. Pattern Anal. Mach. Intell. (1) (1987) 74–90.
[6] G. Franceschetti, V. Pascazio, G. Schirinzi, Iterative homomorphic technique for speckle reduction in synthetic-aperture radar imaging, J. Opt. Soc. Am. A 12 (4) (1995) 686–694.
[7] J.-S. Lee, Speckle analysis and smoothing of synthetic aperture radar images, Comput. Graph. Image Process. 17 (1) (1981) 24–32.
[8] D.T. Kuan, A.A. Sawchuk, T.C. Strand, P. Chavel, Adaptive noise smoothing filter for images with signal-dependent noise., IEEE Trans. Pattern Anal. Mach. Intell. (2) (1985) 165–177.
[9] M. Kuwahara, K. Hachimura, S. Eiho, M. Kinoshita, Processing of RI-angiocardiographic images, in: Digital Processing of Biomedical Images, Springer, 1976, pp. 187–202.
[10] A. Ozcan, A. Bilenca, A.E. Desjardins, B.E. Bouma, G.J. Tearney, Speckle reduction in optical coherence tomography images using digital filtering, J. Opt. Soc. Am. A 24 (7) (2007) 1901–1910.
[11] D.C. Adler, T.H. Ko, J.G. Fujimoto, Speckle reduction in optical coherence tomography images by use of a spatially adaptive wavelet filter, Opt. Lett. 29 (24) (2004) 2878–2880.
[12] M.A. Mayer, A. Borsdorf, M. Wagner, J. Hornegger, C.Y. Mardin, R.P. Tornow, Wavelet denoising of multiframe optical coherence tomography data, Biomed. Opt. Express 3 (3) (2012) 572–589.
[13] P. Puvanathasan, K. Bizheva, Speckle noise reduction algorithm for optical coherence tomography based on interval type II fuzzy set, Opt. Express 15 (24) (2007) 15747–15758.
[14] S. Chitchian, M.A. Fiddy, N.M. Fried, Denoising during optical coherence tomography of the prostate nerves via wavelet shrinkage using dual-tree complex wavelet transform, J. Biomed. Opt. 14 (1) (2009) 014031.
[15] Z. Jian, L. Yu, B. Rao, B.J. Tromberg, Z. Chen, Three-dimensional speckle suppression in optical coherence tomography based on the curvelet transform, Opt. Express 18 (2) (2010) 1024–1032.
[16] H.M. Salinas, D. Cabrera Fernández, Comparison of PDE-based nonlinear diffusion approaches for image enhancement and denoising in optical coherence tomography, IEEE Trans. Med. Imaging 26 (6) (2007) 761–771.
[17] D. Cabrera Fernández, H.M. Salinas, C.A. Puliafito, Automated detection of retinal layer structures on optical coherence tomography images, Opt. Express 13 (25) (2005) 10200–10216.
[18] J. Weickert, Anisotropic Diffusion in Image Processing, vol. 1, Teubner Stuttgart, 1998.
[19] R. Bernardes, C. Maduro, P. Serranho, A. Araújo, S. Barbeiro, J. Cunha-Vaz, Improved adaptive complex diffusion despeckling filter, Opt. Express 18 (23) (2010) 24048–24059.
[20] P. Rodrigues, R. Bernardes, 3-D adaptive nonlinear complex-diffusion despeckling filter, IEEE Trans. Med. Imaging 31 (12) (2012) 2205–2212.
[21] P. Puvanathasan, K. Bizheva, Interval type-II fuzzy anisotropic diffusion algorithm for speckle noise reduction in optical coherence tomography images, Opt. Express 17 (2) (2009) 733–746.
[22] J. Duan, Y. Ding, Z. Pan, J. Yang, L. Bai, Second order mumford-shah model for image denoising, in: International Conference on Image Processing (ICIP 2015), 2015.
[23] J. Duan, C. Tench, I. Gottlob, F. Proudlock, L. Bai, Optical coherence tomography image segmentation, in: International Conference on Image Processing (ICIP 2015), 2015.
[24] L.A. Vese, S.J. Osher, Modeling textures with total variation minimization and oscillating patterns in image processing, J. Sci. Comput. 19 (1–3) (2003) 553–572.
[25] K. Bredies, K. Kunisch, T. Pock, Total generalized variation, SIAM J. Imaging Sci. 3 (3) (2010) 492–526.
[26] X.-C. Tai, J. Hahn, G. Jason Chung, A fast algorithm for Euler's elastica model using augmented Lagrangian method, SIAM J. Imaging Sci. 4 (1) (2011) 313–344.
[27] W. Zhu, X.-C. Tai, T. Chan, Augmented Lagrangian method for a mean curvature based image denoising model, Inverse Probl. Imaging 7 (4) (2013) 1409–1432.
[28] J. Duan, W. Lu, G. Wang, Z. Pan, L. Bai, Second order variational model for image decomposition using the split Bregman algorithm, in: International Conference on Intelligence Science and Big Data Engineering, 2015.
[29] T. Goldstein, S. Osher, The split Bregman method for L1-regularized problems, SIAM J. Imaging Sci. 2 (2) (2009) 323–343.
[30] J. Duan, C. Tench, I. Gottlob, F. Proudlock, L. Bai, New variational image decomposition model for simultaneously denoising and segmenting optical coherence tomography images, Phys. Med. Biol. (2015) (in press).
[31] J. Yang, W. Yin, Y. Zhang, Y. Wang, A fast algorithm for edge-preserving variational multichannel image restoration, SIAM J. Imaging Sci. 2 (2) (2009) 569–592.
[32] L.A. Zadeh, Fuzzy sets, Inf. Control 8 (3) (1965) 338–353.
[33] M. Bashkansky, J. Reintjes, Statistics and reduction of speckle in optical coherence tomography, Opt. Lett. 25 (8) (2000) 545–547.
[34] L.I. Rudin, S. Osher, E. Fatemi, Nonlinear total variation based noise removal algorithms, Phys. D: Nonlinear Phenom. 60 (1) (1992) 259–268.
[35] J. Duan, Z. Pan, B. Zhang, W. Liu, X.-C. Tai, Fast algorithm for color texture image inpainting using the non-local CTV model, J. Glob. Optim. (2015) 1–24.
[36] J. Duan, Z. Pan, X. Yin, W. Wei, G. Wang, Some fast projection methods based on Chan–Vese model for image segmentation, EURASIP J. Image Video Process. 2014 (1) (2014) 1–16.